

Pilot-Aided Channel Estimation Scheme for NR-V2X Speed Emulation Technique

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Abstract—Growing intelligent transportation systems demand a vehicular communication technology that can satisfy high requirements in terms of data rates, latency, reliability and number of connected devices. To evaluate the performance of such communication technology, real-world measurements are required for various channel conditions. Since vehicular measurement campaigns are expensive and time-consuming, a high-mobility environment poses enormous challenges for performance measurements. Using the existing technique of time-stretching the transmit signals, such experiments can be emulated through measurements at a single lower velocity by inducing effects caused by higher velocities. The existing time-stretching technique poses the problem of different channel estimation quality between the time-stretched and the original system. To ensure that the technique gives accurate results in practical systems, we adapt the pilot-based channel estimation scheme within the existing time-stretching technique. Furthermore, we evaluate the proposed channel estimation scheme through simulations and a high-speed vehicular channel measurement campaign at the center frequency of 2.55 GHz.

Index Terms—channel estimation, 5G NR, NR-V2X, high-speed train, physical (PHY) layer, channel measurements.

I. INTRODUCTION

Over the last few decades, the demand for transportation has rapidly increased. Such an increasing demand causes problems in terms of traffic accidents, congestion and greenhouse gas (GHG) emissions [1]. Fortunately, many vehicular applications such as connected autonomous driving, smart navigation and intelligent intersection control have been developed as promising solutions to these problems [2], [3]. In the future, there will be more advances in intelligent transportation systems (ITS) to improve their quality and safety. To support the above mentioned applications, communication systems will need to ensure very high reliability and low end-to-end latency [4]. One of the promising vehicular communication technologies for reaching high reliability and low latency is the fifth generation (5G) new radio (NR) vehicle-to-everything (V2X). V2X communication is characterized by a dynamic environment and high mobility. These characteristics make propagation and channel modeling particularly challenging [5]. To effectively support the design and development of NR-V2X systems, it is crucial to perform channel measurements in high-mobility scenarios.

Performing V2X measurements at different velocities with experimental user equipment is very challenging since it is expensive and time-consuming. Due to high velocities in highway or high-speed train (HST) scenarios, V2X measurements

can sometimes be dangerous or even impossible to perform. As a possible solution, the authors of [6] proposed a time-stretching technique to induce effects caused by highly time-varying channels in orthogonal frequency-division multiplexing (OFDM) signals while conducting measurements at much lower speeds. Thereby, the cost and the time consumption of V2X measurement campaigns can be significantly reduced.

According to [7], the proposed time-stretching technique poses an auxiliary unwanted side effect, if pilot-based channel estimation is employed. More precisely, the time-stretching decreases the pilot spacing in the frequency domain. Therefore, such an effect leads to different channel estimation quality between the time-stretched and the original system. Consequently, the time-stretching technique used within pilot-based channel estimation does not give accurate results in practical systems. Thus, the time-stretched and the original system can not be compared in a fair manner.

Contribution: We propose a novel pilot-based channel estimation scheme for time-stretched systems to ensure a fair comparison between the time-stretched and the original system. Furthermore, we evaluate the proposed channel estimation scheme by conducting simulations and an indoor wireless channel measurement campaign. The measurements are conducted at the center frequency of 2.55 GHz and for different transmitter velocities (50, 100, 150 and 200 km/h) using the testbed hardware and the methodology proposed in [8] and [9].

Notation: Bold uppercase letters denote matrices such as $\mathbf{X}$ and bold lower case letters denote vectors such as $\mathbf{x}$. The size of identity matrix $\mathbf{I}$, all-ones matrix $\mathbf{1}$, all-zeros matrix $\mathbf{0}$ and DFT matrix $\mathbf{F}$ is expressed via their subscripts. The element-wise division of matrices is denoted via the operator $\oslash$. We use the superscript T for transposition.

II. TIME-STRETCHING TECHNIQUE

Let us consider an OFDM communication system within a time-variant multipath propagation channel. A block of K symbols modulated by Quadrature Amplitude Modulation (QAM) is mapped onto K different subcarriers to construct an OFDM symbol. A single OFDM symbol in the frequency domain is represented as vector $\mathbf{x} \in \mathbb{C}^{K \times 1}$, where

$$\mathbf{x} = [x_1 \quad x_2 \quad \dots \quad x_K]^{\text{T}}. \quad (1)$$

Due to channel variation over time, the Inter-Carrier Interference (ICI) arises in the received signal. The amount of the ICI

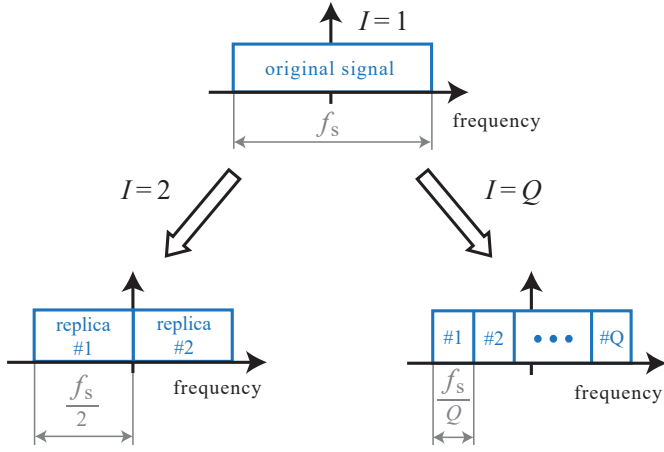


Fig. 1. The original signal is resampled and repeated I times in the frequency domain to occupy the same bandwidth as the original signal.

is related to the normalized Doppler spread D_n of the channel. The normalized Doppler spread is given by $D_n = f_{D_{\max}} \cdot T$, where $f_{D_{\max}}$ represents the maximum Doppler frequency and T represents the symbol duration. As proposed in [6], parameter T can be adjusted by time interpolation by a factor I , yielding a symbol duration $T^I = I \cdot T$. Consequently, an interpolation by a factor I corresponds to a reduction of the subcarrier spacing Δf by the same factor I , given by

$$\Delta f^I = \frac{1}{T^I} = \frac{1}{I \cdot T} = \frac{1}{I} \frac{f_s}{N_{\text{FFT}}}, \quad (2)$$

where f_s represents sampling rate and N_{FFT} represents FFT length. By taking the proposed time interpolation into account, the normalized Doppler spread D_n can be rewritten as:

$$D_n^I = f_{D_{\max}} \cdot T^I = \frac{f_c \cdot v}{c} \cdot I \cdot T = \frac{T \cdot f_c}{c} \cdot v^I \quad (3)$$

where f_c represents the carrier frequency of a signal, c denotes the speed of light and $v^I = I \cdot v$ represents the emulated speed as a result of an actual measurement speed v . As shown in (3), by enlarging the symbol length T^I by an interpolation factor I , it is possible to emulate a velocity v^I while conducting measurements at an actual speed v [10].

One possible method to perform time-stretching is to decrease the sampling rate f_s of given OFDM signal by a factor of I . The decrease of the sampling rate or resampling needs to be done at both the transmitter and the receiver, while FFT size N_{FFT} remains unchanged. Thereby, the subcarrier spacing is decreased by a factor of I compared to the subcarrier spacing of the original signal. Consequently, decreasing the subcarrier spacing Δf by a factor of I decreases the signal bandwidth B by the same factor. To compare the resampled signal with the original one in a fair manner, with respect to the frequency selectivity of the channel, they need to occupy the same total bandwidth B . Therefore, the resampled signal with the bandwidth $\frac{B}{I}$ is repeated I times in the frequency domain. Such a transformation is shown in Fig. 1 and the modified

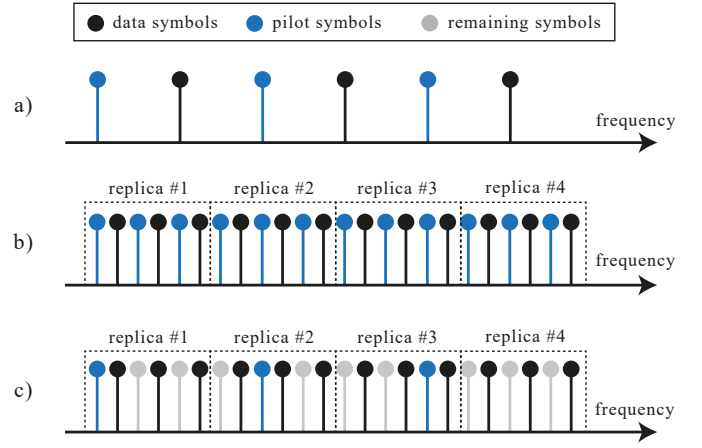


Fig. 2. Pilot insertion schemes for $I = 4$: a) original signal, b) time-stretched signal - original pilot scheme, c) time-stretched signal - proposed pilot scheme

OFDM signal in the frequency domain can be represented as vector $\tilde{\mathbf{x}} \in \mathbb{C}^{KI \times 1}$, given by

$$\tilde{\mathbf{x}} = \begin{bmatrix} (\mathbf{x}^{(1)})^T & (\mathbf{x}^{(2)})^T & \dots & (\mathbf{x}^{(I)})^T \end{bmatrix}^T, \quad (4)$$

where $\mathbf{x}^{(i)}$ represent i -th replica ($i = 1, \dots, I$) of the OFDM signal $\mathbf{x}$.

III. CHANNEL ESTIMATION

In mobile communications, the wireless channel is usually not known at the receiver. Therefore, the channel needs to be estimated to counteract multipath propagation effects. One simple and popular channel estimation method in OFDM systems is pilot-aided channel estimation. Pilot symbols, which are well-known at the receiver, are inserted between data symbols in both the time and frequency domain. These pilot symbols enable the estimation of channel coefficients at pilot positions. After that, channel coefficients at data positions are estimated through interpolation.

A. Original Version

Let us consider the pilot insertion scheme in the frequency domain. The single OFDM symbol consisting of pilot and data symbols is shown in Fig. 2a. After pilot and data mapping, the OFDM symbol $\mathbf{x}$ is resampled and repeated, as explained in Section II. Resampling and repeating the OFDM signal lead to a changed arrangement of symbols, as shown in Fig. 2b. More precisely, the pilot spacing in the frequency-domain decreases with an interpolation factor I . Apart from the decrease of the pilot spacing, the number of pilot symbols is increased by the factor I when repeating the original signal I times in the frequency domain. Such unwanted spectral side effect improves the channel estimation quality in a multipath propagation channel with increasing the factor I . Therefore, the original system (shown in Fig. 2a) and the time-stretched system (shown in Fig. 2b) cannot be compared in a fair manner with respect to the channel estimation quality.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Carrier Frequency	$f_c = 2.55$ GHz
Symbol Alphabet	64-QAM
Number of Subcarriers	$K = 216$
Number of Symbols	$N = 56$
Sampling Rate	$f_s = 20.48$ MHz
FFT Size	$N_{\text{FFT}} = 1024$
Bandwidth	$B = 4.32$ MHz
Interpolation Factor	$I = 1, 2, 3, 4$
Channel Power Delay Profile (Simulation)	TDL-B
Channel Doppler Model	Jakes
Pilot Pattern	Rectangular
Interpolation Method	Linear

B. Proposed Version

To maintain a fair comparison of the original system and the time-stretched system, we adapt the pilot-based channel estimation scheme within the proposed time-stretching technique in the following way. Instead of inserting the pilots before repeating the signal in the frequency domain, this scheme inserts pilot symbols after the signal has been repeated. Firstly, we map only the data symbols onto OFDM subcarriers. Thereby, we leave pilot positions unused. After that, the OFDM signal is repeated I times in the frequency domain, as described in Section II. Secondly, pilot symbols are inserted onto unused subcarriers with the following remarks:

- the spacing between pilot symbols of the time-stretched system in the frequency domain is increased by the factor of I
- the number of pilot symbols of the time-stretched system equals the number of pilot symbols of the original system.

Therefore, frequency-domain pilot spacing and the number of pilot symbols are preserved and correspond to the pilot parameters of the original signal. The proposed scheme is illustrated in Fig. 2c, where a difference to the previously used pilot insertion scheme (shown in Fig. 2b) can be noticed.

In Fig. 2c, there are also remaining symbols that are intended to be used as pilot symbols, as in Section III-A. In the proposed scheme, we consider the remaining symbols to be random symbols that use the same alphabet as the pilot symbols. The main role of remaining symbols is to mimic the ICI, as the receiver is moving at velocity v^I while conducting measurements at an actual speed v . Therefore, such symbols should not be equal to zero. Thus, the adapted scheme opens up the possibility of using remaining symbols for other purposes.

IV. SIMULATION-BASED COMPARISON

For the evaluation of the adapted pilot-based channel estimation scheme, we performed simulation-based comparisons in terms of the channel estimation and interpolation mean square error (MSE) and the bit error ratio (BER) using the Vienna 5G Link Level Simulator [11]. The simulation parameters are given in Table I.

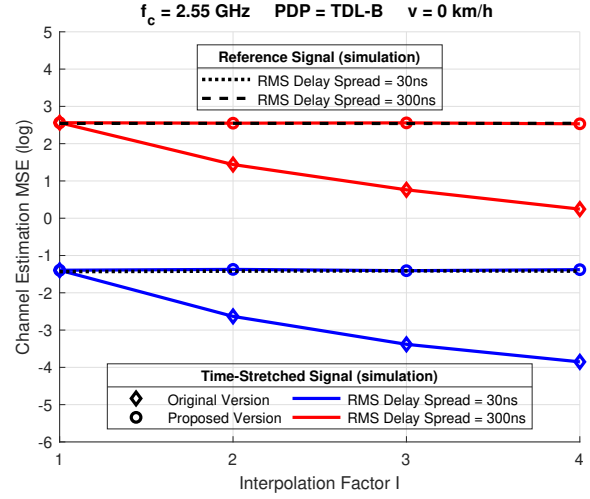


Fig. 3. Using the adapted pilot-based channel estimation scheme, the original frequency-domain pilot spacing is preserved and the time-stretched system can be compared with the original system in a fair manner.

A. Mean Square Error

Firstly, we evaluate the channel estimation and interpolation MSE averaged over $R^s = 1000$ different channel realizations

$$\text{MSE} = \frac{1}{R^s} \frac{1}{N} \frac{1}{K} \sum_{r=1}^{R^s} \sum_{n=1}^N \sum_{k=1}^K |\hat{H}_{k,n,r} - H_{k,n,r}|^2. \quad (5)$$

In (5), $\hat{H}_{k,n,r}$ denotes the estimated channel coefficient and $H_{k,n,r}$ is the actual channel coefficient at subcarrier k , time symbol n and channel realization r . As a reference, we use the MSE for Tapped Delay Line (TDL) B profile, when no time-stretching method is employed. We evaluate multipath propagation channels with root-mean-square (RMS) delay spreads of 30 ns and 300 ns. The simulation results are shown in Fig. 3.

When the pilot-scheme with decreased pilot spacing (original version described in Section III-A) is employed, one can observe that the MSE decreases with interpolation factor I . In other words, the channel estimation quality improves with interpolation factor I . As explained before, it occurs due to the larger number and denser distribution of pilot symbols in the frequency domain. On the other hand, when using the adapted pilot-based channel estimation scheme (proposed version described in Section III-B), one can observe constant MSE performance over the interpolation factor I . That means, there is no difference in channel estimation quality when compared with the TDL-B reference without the time-stretching method employed, which was the goal of this method.

B. Bit Error Ratio

Secondly, we perform simulations in terms of the BER without channel coding employed to prove the applicability of the adapted channel estimation scheme. As a reference, we use an OFDM waveform transmitted over a time-varying channel with the TDL-B delay profile as the transmitter moves at actual

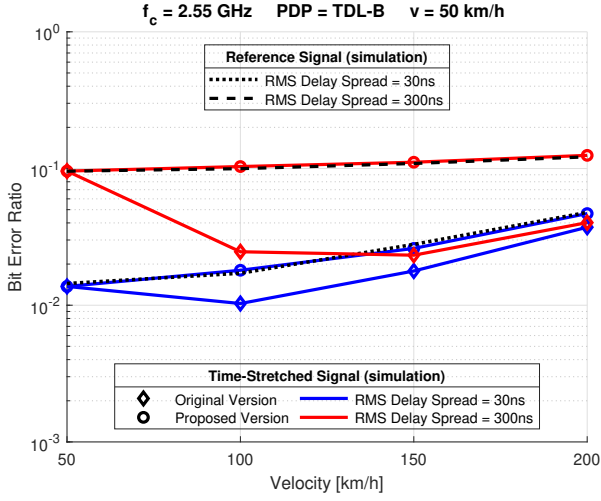


Fig. 4. Using the adapted pilot-based channel estimation scheme, the time-stretched system can be compared with the original system.

velocities of $v = 50, 100, 150$ and 200 km/h. This reference intends to show the signal behavior in the real wireless channel. Then, time-stretched versions of OFDM signals are transmitted as the transmitter moves at the actual velocity of $v = 50$ km/h to emulate transmissions at $v = 100, 150$ and 200 km/h. These emulated velocities correspond to interpolation factors of $I = 2, 3$ and 4 . At the receiver, the BER is calculated as the ratio of erroneously received bits and the total number of transmitted bits. Furthermore, the BER is averaged over $R^s = 1000$ different channel realizations. Simulation results in terms of the BER are shown in Fig. 4.

For the case of the time-stretched system, when the original pilot-scheme is employed, one can observe that the BER does not coincide with its desired reference, simulated for actual transmitter velocity. In other words, the time-stretching technique does not give accurate emulation results. As explained above, this occurs due to the larger number and denser distribution of pilot symbols in the frequency domain. On the other hand, the BER matches with its desired reference when using the proposed pilot-based channel estimation scheme. Furthermore, decreased pilot spacing within the original pilot-scheme impacts the BER more significantly for the channel with a larger RMS delay spread. This occurs due to the stronger frequency selectivity at the RMS delay spread of 300 ns compared to 30 ns.

V. MEASUREMENT-BASED COMPARISON

In addition to simulations, we perform a measurement-based evaluation of the adapted pilot-based channel estimation scheme. We conduct channel sounding measurements using parameters from Table II and Table III. Moreover, we used the channel-sounding results within the link-level simulations (as in Section IV) to prove the applicability of the adapted channel estimation scheme in terms of the BER.

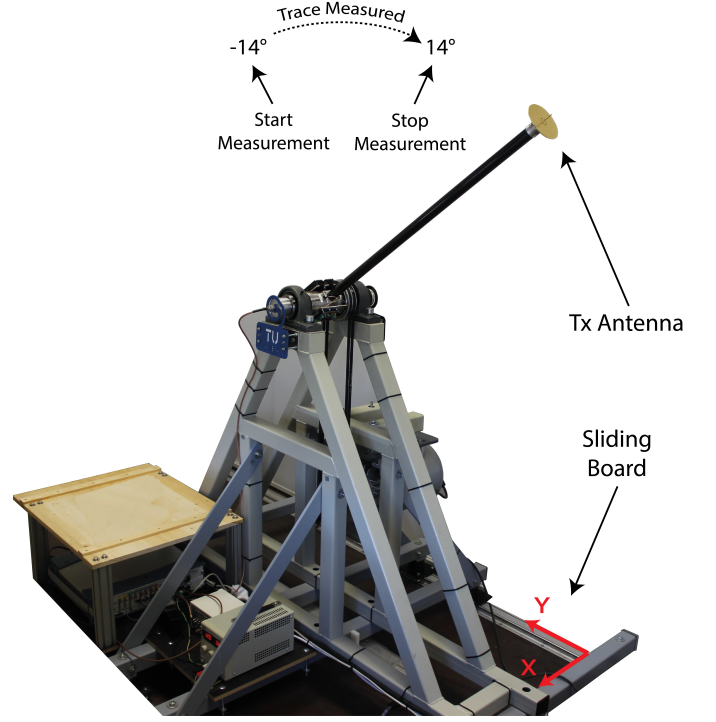


Fig. 5. The rotary unit enables repeatable measurements in a controlled environment at high velocities.

A. Measurement Setup

Our measurement setup is described in [8]. It consists of a moving transmitter equipped with a rotary unit (see Fig. 5) and a static receiver placed in an indoor laboratory environment, shown in Fig. 6. The rotary unit rotates a monopole antenna around a central axis at a constant velocity. This setup enables performing repeatable measurements in a controlled indoor environment at high velocities of up to 400 km/h [12].

B. Channel Sounding

As a transmit signal, we employ the OFDM signal designed using the Zadoff-Chu sequence [13]. A single OFDM symbol is represented as

$$\mathbf{x} = [x_1 \ x_2 \ \dots \ x_K]^T \quad (6)$$

with $x_k = e^{-j\pi \frac{k^2}{K}}$, where k and K represent the subcarrier index and the number of subcarriers, respectively. We transmit N_L identical OFDM symbols as a channel-sounding signal. We performed measurements at the center frequency of 2.55 GHz and for velocities of $50, 100, 150$ and 200 km/h. Channel-sounding parameters that are common for all velocities are given in Table II, while velocity-dependent parameters are given in Table III.

To compare measurements performed at different velocities in a fair manner, we use the methodology proposed in [9]. Firstly, we keep the fading environment static by conducting measurements within 2 hours with no people or objects moving inside the room during the measurement campaign. Secondly, we adjust the symbol duration (the subcarrier

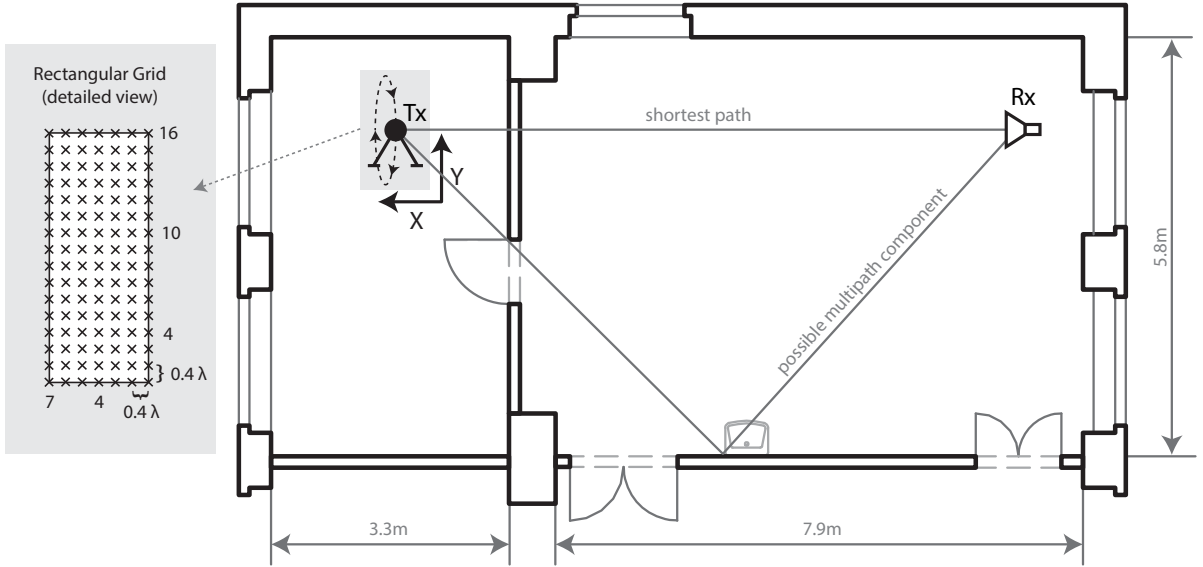


Fig. 6. The moving transmitter and the static receiver are placed in neighboring rooms in an indoor laboratory environment.

TABLE II
COMMON CHANNEL SOUNDING PARAMETERS

Parameter	Value
Carrier Frequency	$f_c = 2.55$ GHz
Wavelength	$\lambda = 11.76$ cm
Bandwidth	$B = 4.32$ MHz
Sampling Rate	$f_s = 300$ Ms/s
Number of Snapshots	$N_{\text{snap}} = 56$
Symbols per Snapshot	$N_{\text{sym}} = 10$
Total Number of Symbols	$N_L = 560$
Start Angular Position	$A_{\text{start}} = -14^\circ$
End Angular Position	$A_{\text{end}} = 14^\circ$
Number of X-axis positions	$N_X = 7$
Number of Y-axis positions	$N_Y = 16$
Number of Realizations	$R^m = 112$

TABLE III
VELOCITY-DEPENDENT PARAMETERS

Parameter	Value			
Transmitter Velocity v [km/h]	50	100	150	200
Number of Subcarriers K	276	138	92	69
FFT Size N_{FFT}	192000	9600	6400	4800
Snapshot Duration t_{snap} [μ s]	640	320	213	160
Measurement Duration t_m [ms]	35.84	17.92	11.95	8.96

2

spacing) to the transmitter's velocity, as given in Table III. Thereby, we ensure transmitting the same number of channel-sounding symbols along the same measured trace for different velocities. The transmission is initiated when the rotating arm passes an angular position of -14° and continues to the angular position of 14° . In addition to the rotation around the central axis, the transmitter can also be moved in space. The transmitter (the rotary unit) is placed on a sliding board, which can be moved by 2.4λ (at 2.55 GHz) along the x-axis and by 6λ along the y-axis. Assuming that the channel is locally stationary for the measured region, we performed

measurements at 112 different positions separated by 0.4λ according to the rectangular grid shown in Fig. 6. Thereby, we obtain $R^m = 112$ different realizations of the same measured wireless channel.

Similar as in [9] and [14], we perform the OFDM processing of the received signal

$$\mathbf{r} = [\mathbf{r}_1^T \quad \mathbf{r}_2^T \quad \cdots \quad \mathbf{r}_{N_L}^T], \quad (7)$$

where $\mathbf{r}_i \in \mathbb{C}^{N_{\text{FFT}} \times 1}$ denotes i -th received signal. Firstly, we group N_L OFDM received symbols into N_{snap} snapshots of N_{sym} symbols each

$$\mathbf{R}_{\text{CP}} = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_{N_{\text{sym}}+1} & \cdots & \mathbf{r}_{(N_{\text{snap}}-1)N_{\text{sym}}+1} \\ \mathbf{r}_2 & \mathbf{r}_{N_{\text{sym}}+2} & \cdots & \mathbf{r}_{(N_{\text{snap}}-1)N_{\text{sym}}+2} \\ \vdots & \vdots & \vdots & \vdots \\ \mathbf{r}_{N_{\text{sym}}} & \mathbf{r}_{2N_{\text{sym}}} & \cdots & \mathbf{r}_{N_{\text{snap}}N_{\text{sym}}} \end{bmatrix}. \quad (8)$$

This grouping is made under the assumption that the wireless channel between the moving antenna and the static receiver is constant in time for the duration of one snapshot. Next, we employ the first OFDM symbol of each snapshot as a cyclic prefix to eliminate Inter-Symbol Interference (ISI) and discard it. After discarding the cyclic prefix, we perform averaging of the remaining $N_{\text{sym}} - 1$ symbols which results in

$$\bar{\mathbf{R}} = [\bar{\mathbf{r}}_1 \quad \bar{\mathbf{r}}_2 \quad \cdots \quad \bar{\mathbf{r}}_{N_{\text{snap}}}], \quad (9)$$

where

$$\bar{\mathbf{r}}_i = \frac{1}{N_{\text{sym}} - 1} \sum_{j=2}^{N_{\text{sym}}} \mathbf{r}_{(i-1)N_{\text{sym}}+j}. \quad (10)$$

Thereby, we improve the signal-to-noise ratio (SNR) by approximately 9.5 dB. Next, we obtain the frequency-domain matrix $\mathbf{Y} \in \mathbb{C}^{K \times N_{\text{snap}}}$ by performing a Discrete Fourier

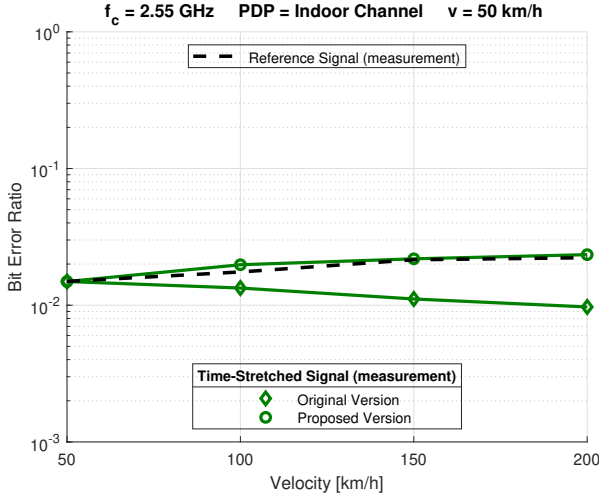


Fig. 7. The adapted pilot-based channel estimation scheme over the measured wireless channel matches its desired reference.

Transform (DFT) and removing the unused subcarriers, in the following way

$$\mathbf{Y} = [\mathbf{I}_K \quad \mathbf{0}_{K \times (N_{\text{FFT}} - K)}] (\mathbf{F}_{N_{\text{FFT}}} \bar{\mathbf{R}}). \quad (11)$$

Finally, we estimate the wireless channel via least-squares estimation for all subcarriers

$$\mathbf{H} = \mathbf{Y} \oslash \mathbf{X}, \quad (12)$$

where $\mathbf{X} = \mathbf{1}_{1 \times N_{\text{snap}}} \otimes \mathbf{x}$ represents the frequency-domain transmit matrix. Thereby, we obtain the time-variant channel transfer function (CTF) for different velocities and for different realizations as the final result of our measurement campaign.

C. Bit Error Ratio

Using the same simulation procedure as in Section IV-B, we evaluate the applicability of the adapted channel estimation scheme. We perform comparisons in terms of the BER using the simulation parameters from Table I and the measured time-varying CTFs instead of TDL-B channels.

The obtained results are shown in Fig. 7. One can observe that the BER within the original pilot scheme does not coincide with its desired reference. Furthermore, decreased pilot spacing from Section III-A does not impact the BER to a great extent due to the small RMS delay spread of the indoor wireless channel. On the other hand, when using the proposed pilot-based channel estimation, the BER matches with its desired reference.

VI. CONCLUSION

The time-stretching technique poses an auxiliary unwanted channel estimation side effect. This effect leads to different channel estimation quality between the time-stretched and the original system. Consequently, the time-stretching technique does not give accurate emulation results in practical systems that utilize channel estimation. We proposed a channel estimation scheme. We conclude that channel estimation within the

time-stretching system is independent of the interpolation factor when done according to our proposed scheme. Therefore, the time-stretching technique with channel estimation leads to the correct performance comparable to the original system and can be used in practical scenarios.

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